Ques. WHAT FACTORS IN CHARGERS AFFECT BATTERY CHARGING SPEED AND HOW DOES MAH DECIDE BATTERY CHARGING?

Syed Muzammil

Ans. Simple battery chargers supply constant direct current (DC) or pulsed DC power source to the battery being charged. Battery charging and discharging speeds are sometimes denoted as C or C-rate, which is a measure of the rate at which a battery is charged or discharged relative to its capacity. C-rate is not a very reliable tool but may be useful to get a rough idea of the battery's maximum discharge current. For example, for a battery with 100mAh capacity, discharge rate of 1000mA (that is, 1A) corresponds to C-rate of 10 (per hour), meaning that such current can discharge ten such batteries in one hour. For the same battery, 50mA charge current corresponds to C-rate of 1/2 (per hour), which means this current will increase the charge state of the battery by 50 per cent in one hour.

Very high charging rates can be used with only a few battery types. Generally, high charging rates are not recommended for normal batteries as the battery may get damaged due to overheating or catch fire. Some batteries may even explode.

Charging protocol depends on the battery size and type. Various factors affect battery charging speed:

Charger's output current. Different chargers come with different output currents. For example, most smartphones require at least 5V, 1A supply, while tablets are charged with 2A or more current. A higher current from the charging device can speed up the battery charging process.

Battery temperature. It plays an important role in battery charging. Heat is the worst enemy of batteries. High battery temperature may slow down charging or reduce lifetime of the battery. Most batteries can be charged in the temperatures range of 5°C to 45°C. However, for best results, battery temperature should be in the range of 10°C to 30°C. The ability to recombine oxygen and hydrogen diminishes when battery (nickel-based) temperature goes below 5°C.

Output quality. This refers to the noise produced by chargers. Good chargers have less noisy outputs than cheap local chargers, which results in faster charging.

Rechargeable batteries—such as those in mobile phones—can store electric energy for a specific period of time (in hours). This capacity of the battery is expressed in terms of milli-ampere hours (mAh). Higher mAh rating means the battery can provide current for a longer time before it completely discharges. Battery capacity decreases as the rate of discharge increases.

A battery can be charged with a current of about 10 per cent of the battery capacity. This can be expressed using the following relationship:

Charging current = Battery capacity (in mAh)/Charging time (in hours)

Let's take the example of a battery with 1200mAh capacity.

Charging current = 1200mAh x (10/100) = 120mA (in ideal case)

Charging time for a 1200mAh battery = 1200/120 = 10 hours

Q2. WHAT IS HAM RADIO? WHAT ARE THE ADVANTAGES OF BECOMING A HAM RADIO OPERATOR?

A. Samiuddhin

A2. Ham radio, also known as amateur radio, is a fun and exciting hobby involving radio transmitters and receivers. There is a large community of ham radio operators across the globe. Ham radio operators come from all walks of life, including professionals, students, politicians, truck drivers, movie stars, missionaries and others. They use radio frequency spectrum for non-commercial communication or messages, wireless experimentation, private recreation, emergency communication and other purposes. This two-way radio service for hobbyists is often very useful in emergency or disaster situations.

Ham radio service is provided under Radio Regulations of the International Telecommunication Unit (ITU). National governments regulate technical and operational characteristics of signal transmissions and issue individual stations licences with an identifying call sign. That is, ham radio operators use a call sign on the air to identify the operator and/or station. Call sign structures are prescribed by the ITU, though some countries have their own signs.

The ITU has divided all countries into three regions. India is located in ITU Region 3. The ITU has assigned following call-sign blocks to India: 8TA to 8YZ, VUA to VWZ, and ATA to AWZ. For example, Indian amateur radio call-signs are VU2XY and VU2XYZ.

For more information on ham radio, you may go through the following websites:

https://en.wikipedia.org/wiki/Amateur_radio_in_India

http://www.hamradio.in/amateur_radio/

http://www.hamradioindia.com/

http://vigyanprasar.gov.in/science-communication-programs/ham-radio/

http://www.cq-amateur-radio.com/

There are many advantages to becoming a ham radio operator. Some of these are listed below:

1. Ham operators can talk to other ham operators located in any part of the globe with transmitter output power of 10 watts or even less
2. When cellphones, Internet and other communication systems are down or overloaded, ham radio still gets the message through.
3. Ham operators provide hours of volunteer community and emergency services during large-scale disasters and natural calamities.
4. They can build their own project or experiment with use of a simple antenna and low-power transmitter or QRP HF radios, and even interface the circuit to a computer.
5. Amateur radio operators on the Earth can communicate directly with astronauts/cosmonauts in a space station via their equipment or home radio stations. It is also possible to send digital data to the space station via laptop computers connected to the same equipment.

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